Davis & Furber Spinning Mule

That's a cool machine! It's also a very old machine. How old? Well, it's hard to tell, but we know it's over **100 years old**! Let's look for clues that can help us narrow it down.

Clue #1: Power Input

The machine you see is being powered by an electric motor hidden in the ceiling right now. But that's not how it was designed. Most Davis and Furber mules that were designed for electric motors had them built-in and hidden in the sides, like this one from 1956:

The machine here doesn't look like that. It instead has large wheels in the back that are turned by a belt:

You don't need to power the machine this way with an electric motor, since you can keep an electric motor's output consistent. So this machine is even older than electric motors, which were popular by the beginning of the 20th century. This machine's power input is more compatible for water wheels instead. Textile machines were starting to phase out water wheel designs and bringing in the electric motor designs by the **late 1880s**, meaning this machine is even older than that!

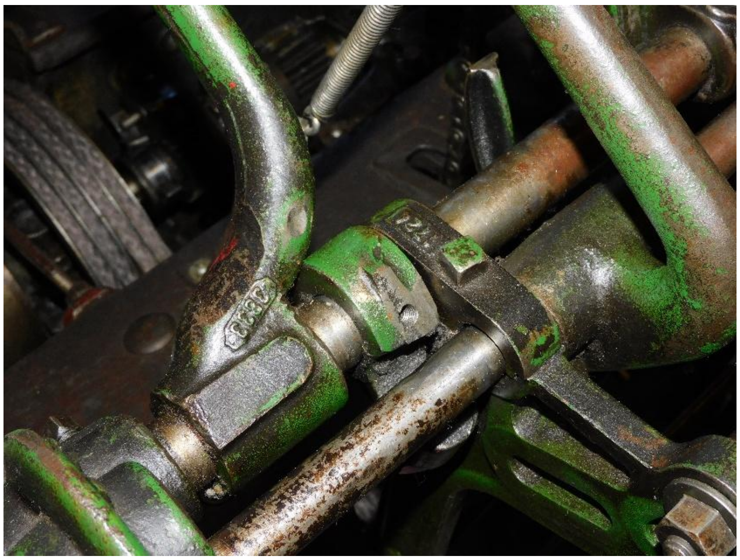
Clue #2: Artistic Details

The machine looks nice right? Especially the wooden details and hand-cast brass emblems on the front. Well, this is also a sign of the times, when each machine coming off the assembly line was mostly hand-made and machine-assisted instead of machine-made with some human input. Check out these details:

The machine is mostly cast iron, which was first used by the ancient Chinese as early as 600 B.C.E. Obviously, this machine isn't quite *that* old, but it's a good starting point. Most machines were made from wood until the industrial revolution, which introduced metal as a stronger alternative to wood. Davis and Furber didn't produce fully cast-iron machines until **about the 1890s,** which can be seen from the carding machine across from the spinning mule.

Clue #3: The industrial capabilities of D&F at the time

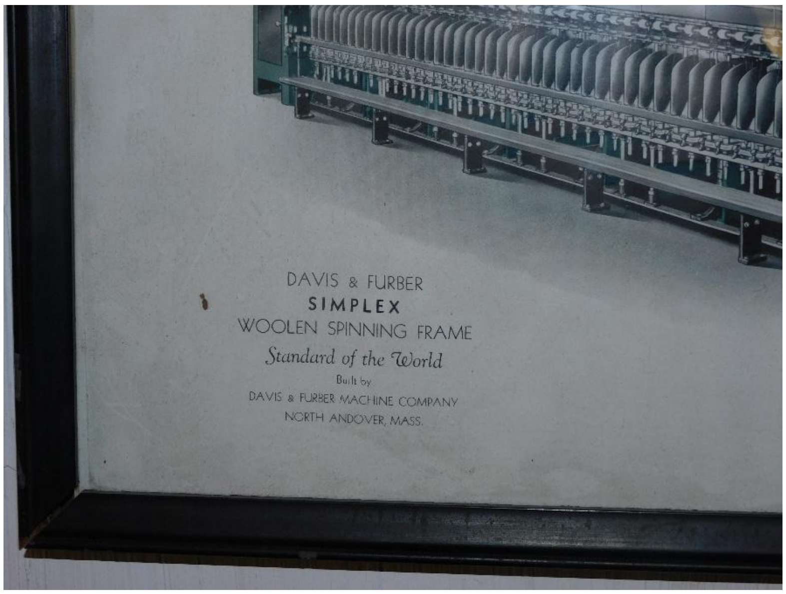
Did you know that this spinning mule is one of only 149 ever built? Not only does that make it extremely rare, but it also helps us figure out when it was made. 149 is an extremely small number for a product line, meaning that this machine was probably made before Davis and Furber had heavy industrial machinery to make parts for these machines quickly. Moreover, all of the parts have part ID numbers on them, which is essentially equivalent to a serial number in today's manufacturing. Take a look at these parts:

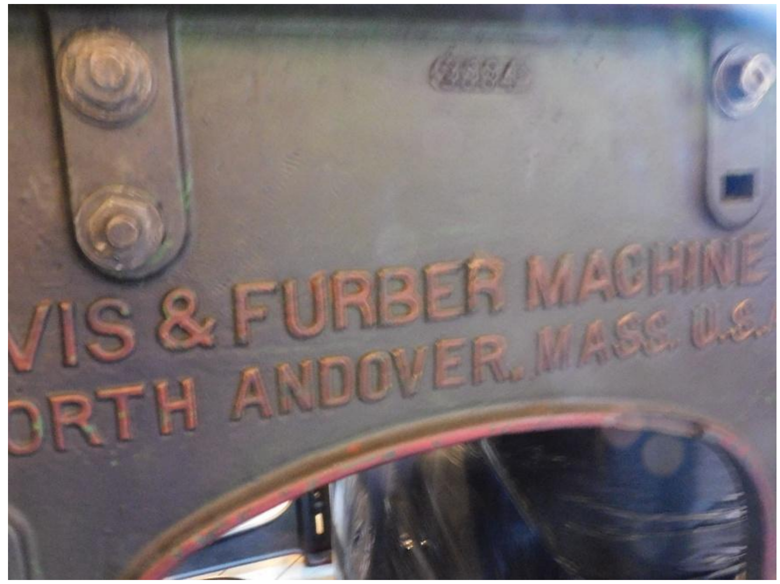


These parts' serial numbers are only **4 digits long**! Most serial numbers for today are at least 12 digits, and include letters on top of that. This indicates a very early production line **no more than 25 years after Davis & Furber's founding**.

Clue #4: Manufacturer

We've talked about it before, but this machine is manufactured by Davis & Furber, a company based in North Andover, Massachusetts, U.S.A. They specialized in reliable machines with easy-to-replace parts, and as a result, some of their machines are able to work for over 300,000 hours before needing to be fully replaced! For comparison: 8 hours a day, 5 days a week, every week since 1956 is just over 150,000 working hours!

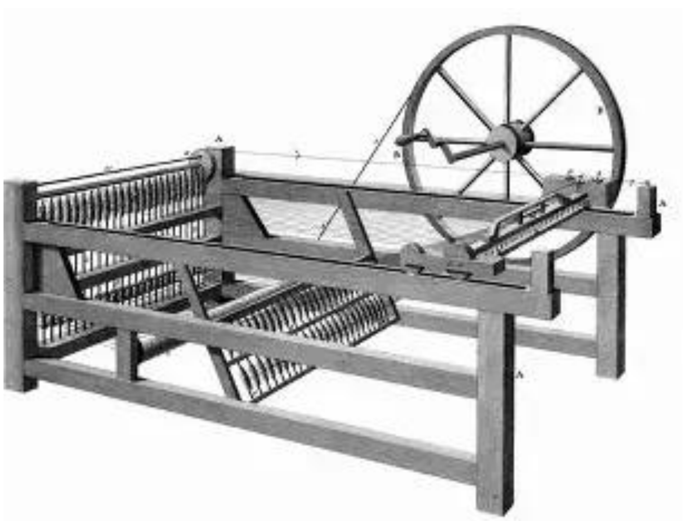


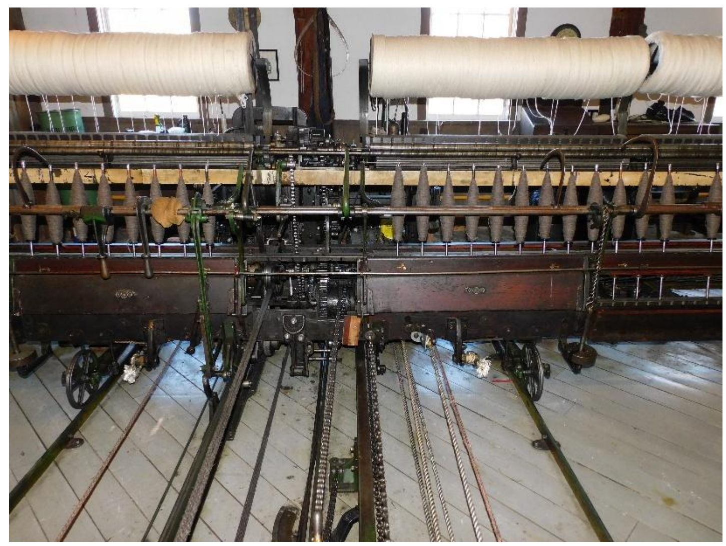


Davis & Furber was **first founded in 1851**, and their peak machining days were **a decade or two later** for machines looking like this one. According to 1936 catalogs, the machines were **fully electrical and fully cast iron** by this point.

Clue #5: Usage of spinning mules

The Ulverton Wool Mill makes yarn out of wool. Obviously, right? Well, this is actually really important to talk about, because the process of spinning wool into yarn needs very different machines from spinning cotton or silk. Woolen spinning has evolved quite a bit over the years, and it all started with the spinning jenny. These machines were popular throughout the 1700s, and they were replaced with spinning mules after their popularization in 1790. Spinning mules were the most popular spinning machine from **1825-1900**, when they were then replaced by ring spinning frames. Since ring spinning frames were becoming popular, companies switched over to making ring spinning frames and no longer made spinning mules. Generally, spinning mules **stopped production in the late 1890s to early 1900s**, to allow new technologies to come into the industry.



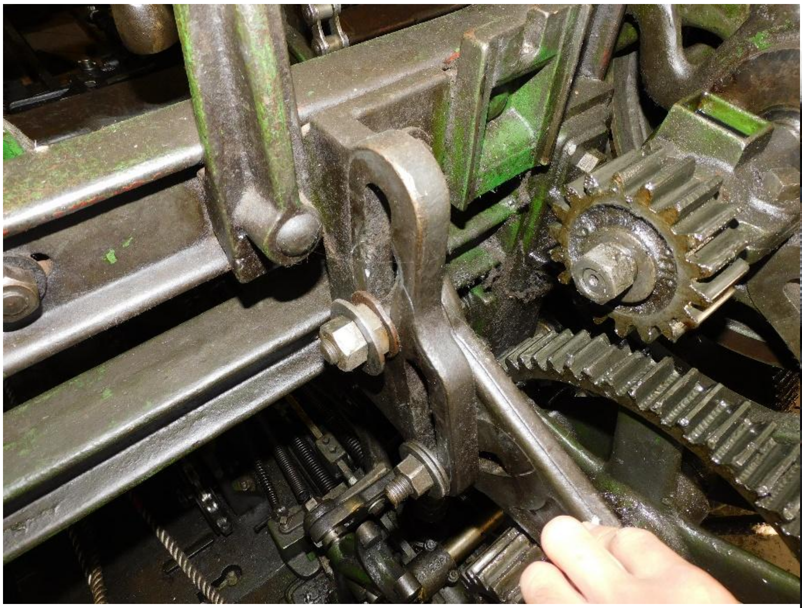


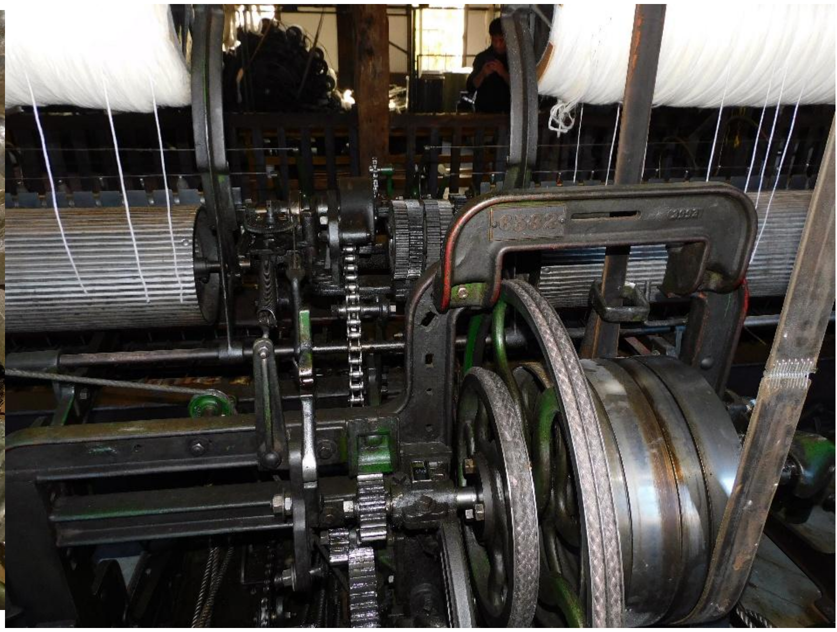
Clue #6: Safety Features

Designing machines, especially complicated ones like the spinning mule, is very difficult. Designing machines to be easily repaired, like the spinning mule, is even harder. Designing machines to be safe to use on top of that is harder still. Because of that, making stuff was exclusive to very skilled people using specialized machines that weren't designed to keep the users safe. After the industrial revolution (**1750-1850**), however, more people were getting into the business of making stuff, so the job wasn't reserved for highly skilled people anymore. Because of that, machines needed to be built while thinking about the risks that using it poses to factory workers.

By the **end of the industrial revolution**, safety was being considered in the design of machines. But, take a look at the spinning mule. There's a lot of exposed gears, a lot of heavy lifting in its operation, and places to get your fingers stuck. This means that **safety regulations weren't quite mature when the machine was designed**, and still requires an operator with proper training to use safely. These safety features were being developed **after the end of the industrial revolution**. But, most designers from the 19th and early 20th centuries only considered safety as a top priority if the machines they were designing **weren't able to improve the effectiveness very much**.

This machine, on the other hand, was built when the margin for improvement was **very high**, so the operator being safe is less of a priority.





Clue #7: Fastening bolts

Take a look at the bolts on the machine. You'll notice that they have two different head shapes: square and hexagonal. These actually come from two different time periods, square bolts were used **before 1900** and hexagonal bolts were used **after 1900**. Both types are made from cast iron, which was replaced by stainless steel for bolts in the **early 1930s**. But why the two types of bolt?

Well, the spinning mule was actually reassembled after the Ulverton Mill closed down. The machine was taken apart and put into storage while the Mill was being traded around. When the heritage museum opened, the machine had to be rebuilt using newer hexagonal bolts. But the original machine was likely put together with **square bolts from before 1900**.